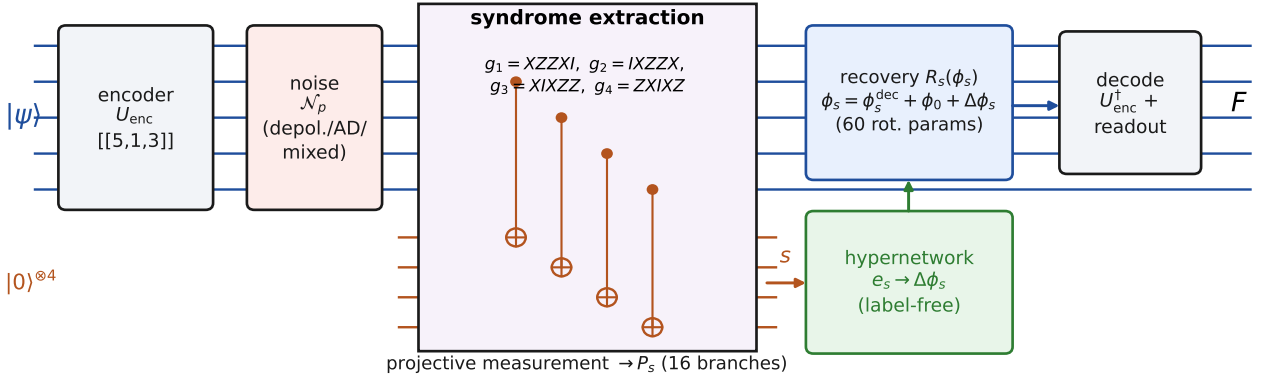
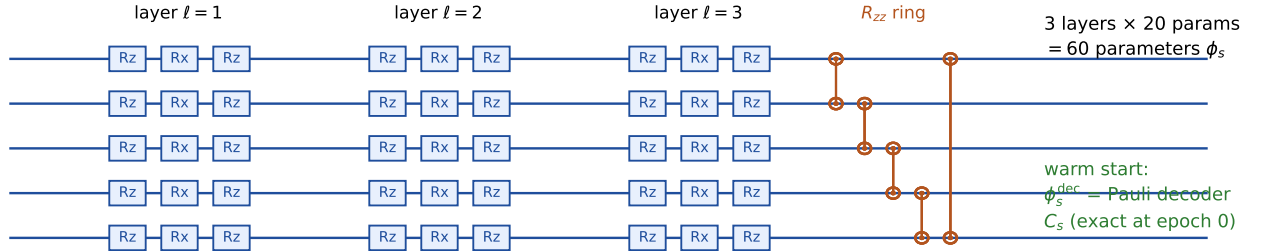


(a) Variational syndrome-conditioned recovery (VSCR)

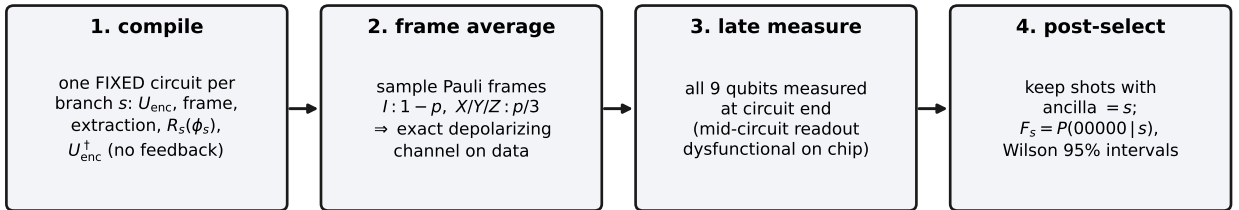


(b) Learned recovery ansatz R_s and warm-start training



unsupervised loss: $\mathcal{L} = 1 - \sum_s |\langle \psi | R_s P_s \mathcal{N}_p[\psi] P_s R_s^\dagger | \psi \rangle|$ (no error labels; Adam + curriculum; multi-seed; label-free selection)

(c) Branch-fixed hardware benchmark on a superconducting QPU



$F(p) = \mathbb{E}_{\text{frames}} \sum_s P(s) F_s$ — hardware estimate vs. exact circuit-level ideal reference (verified $< 10^{-9}$ vs. density-matrix simulator)